SIMATS ENGINEERING

ASSESSMENT TOOL 1 - SHORT ANSWER TEST

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Course Name	Data Warehousing and Data Mining	**Course Code**	1613
Course Outcome	CO1 (BL1, BL2): Design data warehouse architectures with appropriate dimensional models for effective decision support systems	**Total Marks**	20 (4 questions x 5)

Code of Conduct:

I, J O Joanna Divine (Reg No 192572003), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Joanna*

Question 1 (5 Marks)

A data warehouse has three dimensions - time, doctor and patient - and two measures - count and charge (the fee a doctor charges a patient for a visit).

(a) Three classes of schemas used for modeling data warehouses

Schema	Description
1. Star schema	One central fact table linked directly to a set of de-normalized dimension tables (one table per dimension). Simple, fewest joins, fast queries.
2. Snowflake schema	A refinement of the star schema in which dimension tables are normalized into several related tables (e.g. a hierarchy is split into separate tables). Saves space but needs more joins.
3. Fact constellation (galaxy) schema	Multiple fact tables share common dimension tables - a collection of star schemas. Used for sophisticated, enterprise-wide warehouses.

(b) Schema diagram (Star schema)

The fact table **Visit_Fact** holds the two measures (count, charge) and a foreign key to each of the three dimension tables.

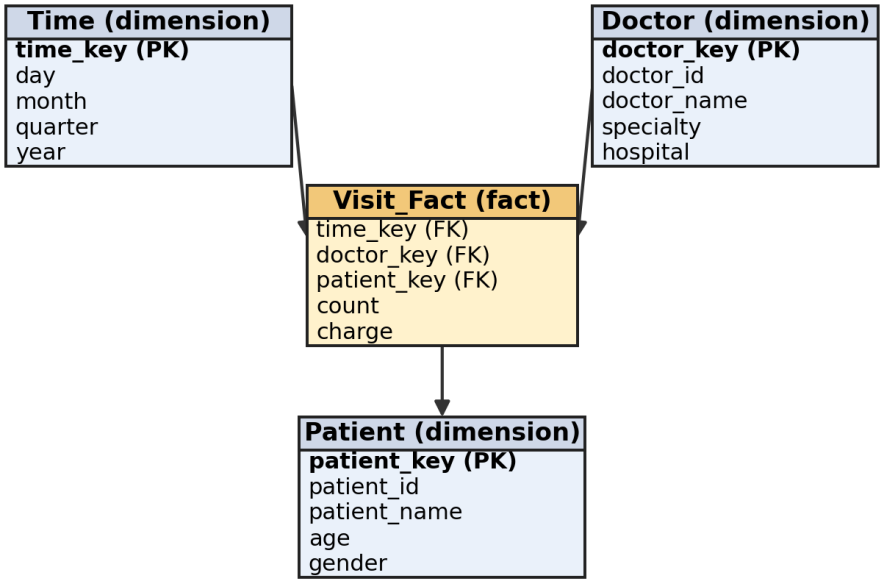


Figure 1: Star schema for the hospital data warehouse (time, doctor, patient).

(c) SQL query

The list asked for is the **total fee collected by each doctor in the year 2010**, computed from the relational table `fee(day, month, year, doctor, hospital, patient, count, charge)`:

```
SELECT   doctor, SUM(charge) AS total_fee
FROM     fee
WHERE    year = 2010
GROUP BY doctor;
```

Equivalent OLAP view: starting from the base cuboid [day, doctor, patient], *roll-up* time from day to year, *slice* on year = 2010, and *roll-up* patient to all - leaving the 1-D cuboid [doctor] with the sum of charge.

Question 2 (5 Marks)

Big University warehouse: dimensions student, course, semester, instructor; measures count and avg_grade.

(a) Snowflake schema diagram

Grade_Fact (measures: count, avg_grade) links to the four dimensions. Each dimension is normalized along its concept hierarchy: student < major < status < university, course < department, semester < year.

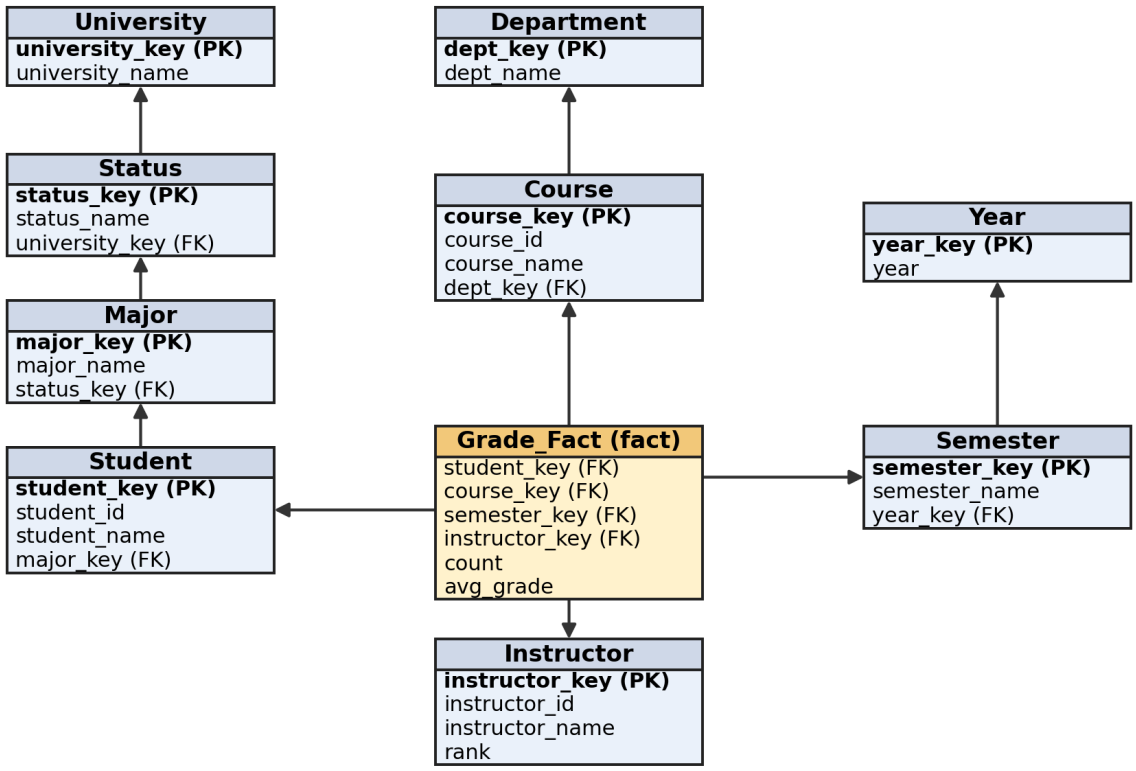


Figure 2: Snowflake schema for the Big University data warehouse.

(b) OLAP operations to list the average grade of CS courses for each student

Starting from the base cuboid [student, course, semester, instructor]:

- **Roll-up** on *course* from course_id to **department**.
- **Roll-up** on *semester* from semester to **all** (ignore the time dimension).
- **Roll-up** on *instructor* from instructor to **all**.
- **Slice / dice** on *course* with department = "CS" (select only CS courses).
- **Student** is kept at the lowest level (student_id), so the result is one row per student.

Result: the cuboid [student, department = CS] giving avg_grade for every student.

(c) Number of cuboids

Each dimension has 5 levels (including *all*), so every dimension can be shown at any of 5 levels. A cuboid is one combination of a level from each of the 4 dimensions:

Number of cuboids = 5 x 5 x 5 x 5 = 5^4 = **625** (including the base cuboid and the apex cuboid).

Question 3 (5 Marks)

Design a data warehouse for a regional weather bureau (about 1,000 probes on land and ocean; hourly air pressure, temperature and precipitation; more than 10 years of data).

Step 1: Choose the subject and granularity

Subject: **weather**. The lowest granularity is **one reading per probe per hour**.

Step 2: Dimensions and concept hierarchies

Dimension	Attributes / concept hierarchy
Time	hour < day < month < season < year < all (also day-of-week, day/night)
Location	probe location (latitude, longitude, altitude) < district < region < all; also land / ocean
Probe	probe_id, probe_type, install_date (to analyse data quality of individual probes)

Step 3: Measures (in the fact table)

- **temperature** (avg, min, max), **air_pressure** (avg, min, max), **precipitation** (sum, max) and a **count** of readings.
- All are numeric and aggregatable, so they support roll-up with sum / avg / min / max.

Step 4: Schema - star schema

A **star schema** is chosen: the central **Weather_Fact** table joins to the Time, Location and Probe dimensions, which keeps joins few and OLAP queries fast.

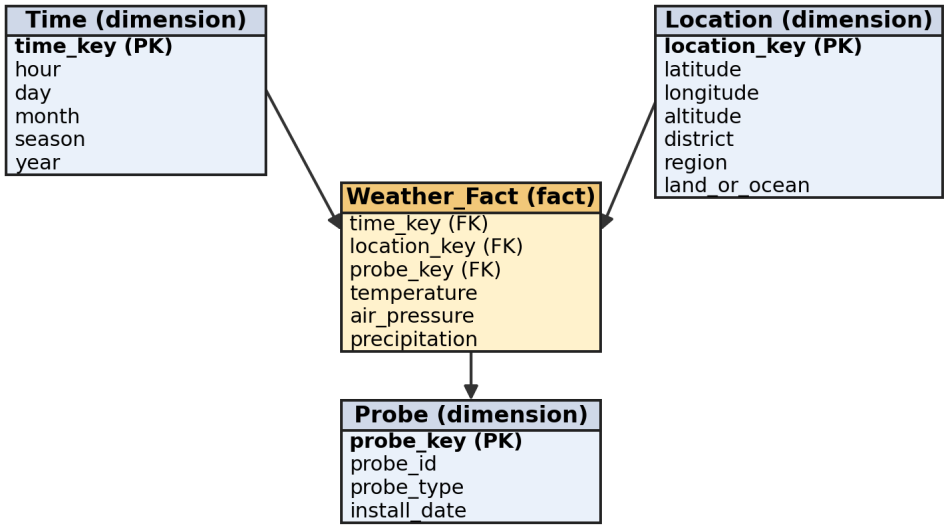


Figure 3: Star schema for the weather bureau data warehouse.

Step 5: Data size estimate

1,000 probes x 24 hours x 365 days x 10 years = **about 87.6 million** base fact records. This is large, so selected aggregate cuboids are **pre-computed (materialized)** for efficient querying, e.g. [day, district], [month, region] and [season, region].

Step 6: OLAP support and pattern discovery

- **Roll-up / drill-down:** hour to day to month to year; probe to district to region.
- **Slice / dice:** a single season, or ocean probes only; a region and a year range.
- **Pivot:** view time x region tables of avg temperature or total precipitation.
- Multidimensional analysis then reveals general weather patterns, e.g. seasonal temperature trends per region, rainfall differences between land and ocean, pressure patterns before heavy rain, and extreme-weather detection.

Question 4 (5 Marks)

Data cube C has n dimensions; each dimension has exactly p distinct values in the base cuboid; there are no concept hierarchies.

(a) Maximum number of cells in the base cuboid

Each of the n dimensions can take any of p values, so the largest possible base cuboid has every combination: $\mathbf{p^n}$ cells.

(b) Minimum number of cells in the base cuboid

Every dimension has exactly p distinct values, and each value must appear in at least one cell. So at least p cells are needed (e.g. the cells $(a_i, a_i, ..., a_i)$ for i = 1..p): **p** cells.

(c) Maximum number of cells in the whole data cube C

With no hierarchies, each dimension is either at its p values or aggregated to *all* (1 extra value). Each dimension therefore has (p + 1) possibilities:

Maximum cells = (p + 1) x (p + 1) x ... (n times) = $\mathbf{(p + 1)^n}$.

(d) Minimum number of cells in the data cube C

Take the minimum base cuboid of p cells arranged as in (b). The cube has 2^n cuboids:

- The apex cuboid (all dimensions aggregated) has **1** cell.
- Each of the other $(2^n - 1)$ cuboids has at least **p** cells, since each remaining dimension still shows all its p values and the cells line up one-to-one.

Minimum cells = $(2^n - 1)$ x p + 1 = $\mathbf{(2^n - 1)p + 1}$.

Part	Quantity	Answer
(a)	Max cells in base cuboid	p^n
(b)	Min cells in base cuboid	p
(c)	Max cells in data cube C	$(p + 1)^n$
(d)	Min cells in data cube C	$(2^n - 1)p + 1$